

HAT-P-32b

R-1285

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_211206 · created 2026-07-31T15:33:53+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459554.6604 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1139 +/- 0.0096
Transit depth (Rp/Rs)^2	1.3 +/- 0.22 [%]
Orbital Inclination (inc)	89.82 +/- 1.63
Airmass coefficient 1 (a1)	1.0651 +/- 0.0088
Airmass coefficient 2 (a2)	-0.054 +/- 0.02
Scatter in the residuals of the lightcurve fit is	0.83 %
Best Comparison Star	None
Optimal Aperture	4.25
Optimal Annulus	10.84
Transit Duration (day)	0.1258 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1139 $\pm$ 0.0096 vs 0.1489 (deviation 3.48 σ)
Duration fit vs published	0.1258 d vs 0.1304 d (deviation 2.59 σ)
Mid-time O-C	-8.8 min
Depth SNR	11.3
Residual scatter	0.83 %

Light curve

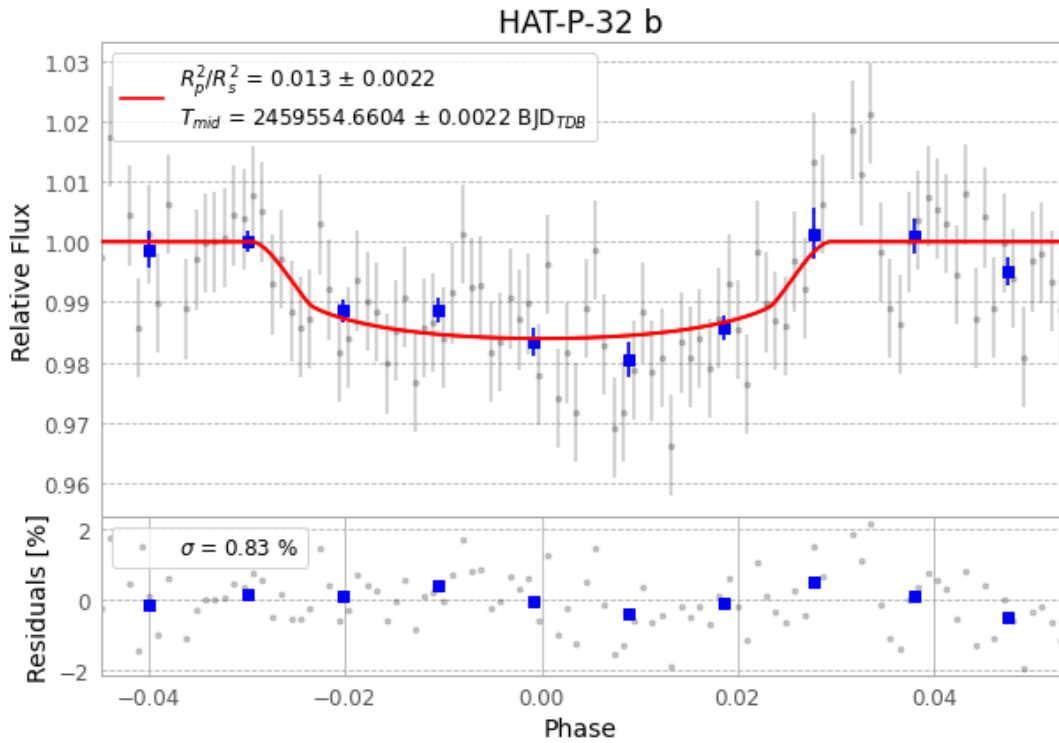


Figure 1 — FinalLightCurve_HAT-P-32 b_2021-12-05.png (SHA-256 a1c7768f80b2d001...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [333, 233], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d69eb6b36143d800...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

100 FITS frames (66 MB), HATP-32211206012412.FITS → HATP-32211206062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-211206.log (c40d6d0738e1...), FinalLightCurve_HAT-P-32 b_2021-12-05.png (a1c7768f80b2...), FinalLightCurve_HAT-P-32 b_2021-12-05.csv (2ca52f199315...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 15:33Z.